

Jacobi's method

Since Jacobi's method is easy to use, I will introduce it through an example, instead of through a formal definition.

Example 1 Use Jacobi's method to get an estimate of the solution of

$$\begin{aligned}5x_1 + x_2 + x_3 &= 1 \\ -x_1 + 4x_2 - 2x_3 &= 1 \\ -2x_1 - 2x_2 + 6x_3 &= 1\end{aligned}$$

Start with an initial estimate (of the solution) of $x = [1, 1, 1]^T$.

Answer: Re-arrange the first equation so that it is of the form $x_1 = \dots$, the second equation so that it is of the form $x_2 = \dots$, and the third equation so that it is of the form $x_3 = \dots$. This is very easy and you get

$$\begin{aligned}x_1 &= (1 - x_2 - x_3)/5 \\ x_2 &= (1 + x_1 + 2x_3)/4 \\ x_3 &= (1 + 2x_1 + 2x_2)/6\end{aligned}$$

We are now ready to perform the iterations.

Iteration 1

We are told the initial estimate is $[1, 1, 1]^T$. So, on the right hand side of the re-arranged equations we use $x_1 = 1$, $x_2 = 1$ and $x_3 = 1$. This gives

$$\begin{aligned}x_1 &= (1 - 1 - 1)/5 = -0.2000 \\ x_2 &= (1 + 1 + 2(1))/4 = 1.0000 \\ x_3 &= (1 + 2(1) + 2(1))/6 = 0.8333\end{aligned}$$

i.e. the approximation to the solution after one iteration is $[-0.2000, 1.0000, 0.8333]^T$ (to four decimal places).

Iteration 2

On the right hand side of the re-arranged equations we use the values of x_1 , x_2 and x_3 from the first iteration i.e. $x_1 = -0.2000$, $x_2 = 1.0000$ and $x_3 = 0.8333$. This gives

$$\begin{aligned}x_1 &= (1 - 1 - 0.8333)/5 = -0.1667 \\ x_2 &= (1 + (-0.2) + 2(0.8333))/4 = 0.6167 \\ x_3 &= (1 + 2(-0.2) + 2(1))/6 = 0.4333\end{aligned}$$

i.e. after the second iteration, we have the approximate solution $[-0.1667, 0.6167, 0.4333]^T$.

More iterations

It is tedious to do the iterations by hand, so I used matlab, and did another 13 iterations. The approximate solution after 5, 10 and 15 iterations is given in the table below.

Iteration	x_1	x_2	x_3
5	0.0578	0.4154	0.3192
10	0.0500	0.4250	0.3250
15	0.0500	0.4250	0.3250

The difference between the approximations for the 10th and 15th iterations is less than the difference between the approximations for the 5th and 10th. This suggests the iterations are converging. Furthermore, since the approximations for the 10th and 15th iterations agree to four decimal places, the solution to the system of equations (to four decimal places) is probably $x_1 = 0.0500$, $x_2 = 0.4250$, $x_3 = 0.3250$.